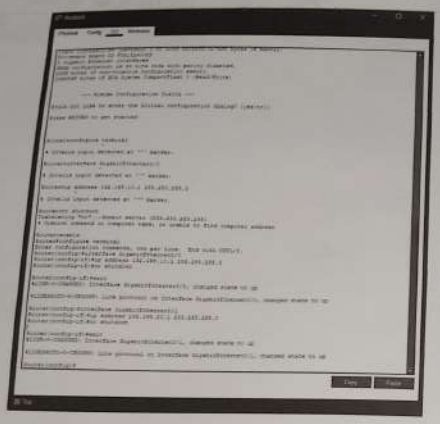
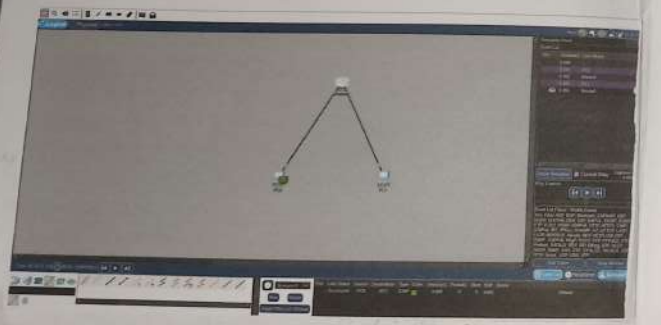
Ex:10 a) Internetworking with Routers.





Aim

To design and configure a simple internetwork using a router in Cisco Packet Tracer simulator and simulate packet transmission between two different networks

Procedure:-

* Open Cisco packet Tracer and create a new network topology
* Add the following
  1. Router - Router0
  2 PCs - pc0 and pc1
* Connect pc0 to Router0 using Copper Straight-Through Cable
* Connect pc1 to Router0 using a Copper Straight-Through cable.
* Open Router0 - CLI and enter privileged mode:-

```
enable
configure terminal
```

* configure the first router

```
interface GigabitEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
```

* configure the second router

```
interface GigabitEthernet0/1
id address 192.168.20.1 255.255.255.0
no shutdown
exit-.
```

* PC0 using Desktop-IP configuration:-
  IP Address 192.168.10.2
  Subnet Mask 255.255.255.0
  Default Gateway : 192.168.10.1

* PC1:
  IP Address : 192.168.20.2
  Subnet Mask : 255.255.255.0
  Default Gateway : 192.168.20.1

* Switch to simulation Mode.
* Select Add simple PDU and click PC0 as the source and PC1 as the destination.
* Use Capture/Forward to trace the packet
* Observe the packet path:
  PC0 → Router0 → PC1
* Continue the simulation and observe the acknowledgement.
  PC1 → Router0 → PC0
* Verify the successful status and the tick mask on PC0.
* Send a PDU from PC0 to PC1
* Verify the communication between the two networks

Result:-
Thus, the internetwork was successfully configured using a routes and connectivity was verified.

Exp.No-10 b

Wireless Internetworking.

Aim :-

To design and configure an internetwork using a wireless router, DHCP server and Internet cloud.

Algorithm :-

* Open cisco packet tracer.
* Add a wireless router, DHCP server and Internet cloud.
* Add the required wireless devices
* Connect the devices to the wireless router
* Configure the wireless router settings.
* Configure the DHCP server with Ip settings
* Enable DHCP for automatic Ip assignment.
* Connect the wireless device to the network
* verify the assigned Ip address.
* Test connectivity between the devices and Internet cloud.

Result :-

Thus, the wireless internetwork was successfully configured and connectivity was verified.